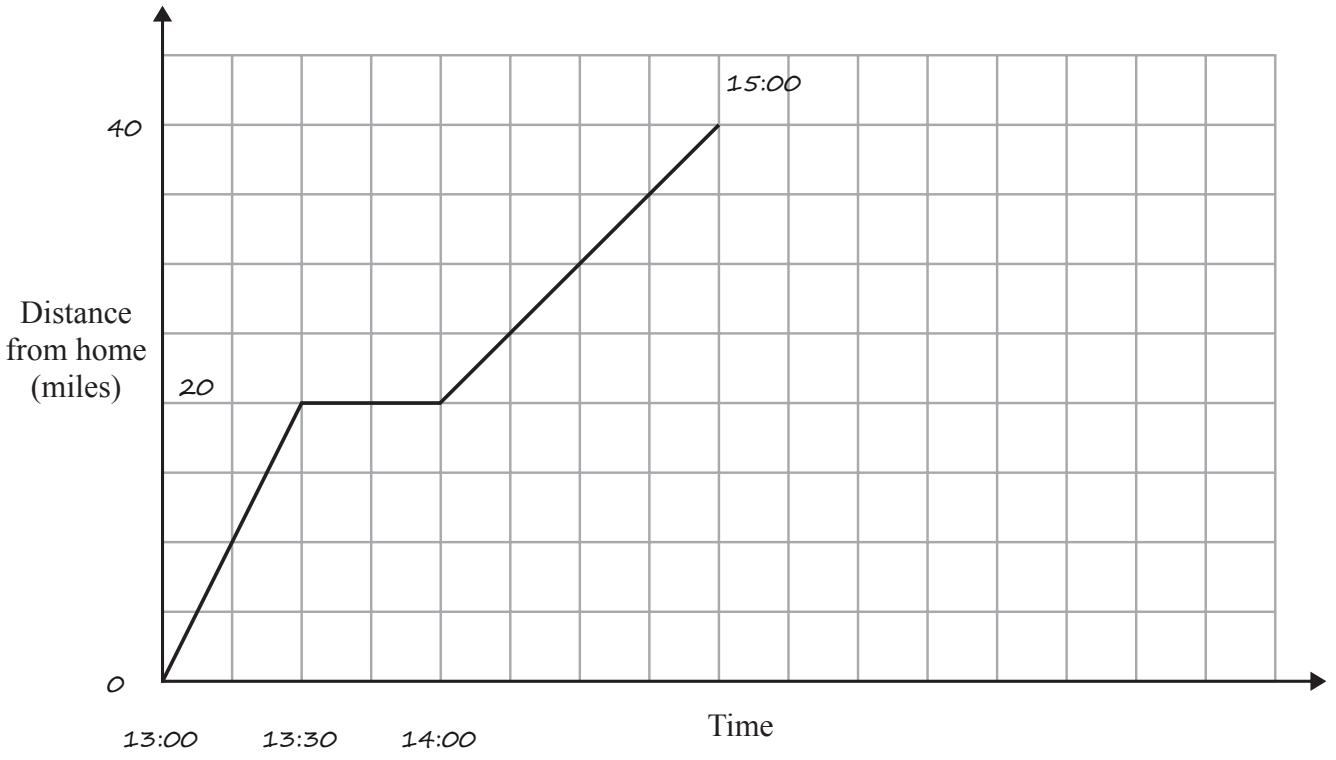
22 On Monday Ria delivered a parcel to a hospital.
The travel graph represents Ria's journey to the hospital.



Ria left home at 13 00
She drove for 30 minutes at a constant speed of 40 mph.
She then stopped for a break.

Ria then drove to the hospital at a constant speed.
She was at the hospital for 30 minutes.
She then drove home at a constant speed of 32 mph.

Show that she does not arrive home before 16 30

hours	miles
1	40
0.5	20

(÷2 and ÷2)

miles	hours
32	1
40	1.25

(×1.25 and ×1.25)

1 mark
First stage 20 miles

1 mark
Hospital 40 miles from graph scale

Hospital at 15:00; leave after 30 min at 15:30.

1 mark
Return takes 1¼ h

Home journey = 40 ÷ 32 = 1.25 h = 1 h 15 min.

15:30 + 1:15 = 16:45, which is after 16:30.

1 mark
Arrives 16:45

(Total for Question 22 is 4 marks)